

SEND5232: Software Architecture and Design Project

Bereket Kiros – ugr/189518/16 – Section 2

Phase 3 - ASRs by Development Perspective

Lihq'et Bookstore System

The Architecturally Significant Requirements (ASRs) of the Lihq'et Bookstore System are classified into four development perspectives: Frontend, Backend, DevOps, and Logic/Business Layer. This classification ensures clear separation of responsibilities, improves system modularity, and aligns with MVC and layered architectural principles.

Frontend Perspective: The frontend ASRs focus on usability, responsiveness, and user interaction, requiring a simple and intuitive interface for managing books, real-time UI updates after CRUD operations, and seamless communication with backend APIs using asynchronous Fetch calls to ensure a smooth user experience.

Backend Perspective: The backend ASRs emphasize performance, security, and maintainability, requiring structured RESTful API design using Express.js, implementation of MVC with a Service Layer to separate business logic, secure handling of user inputs to prevent injection attacks, and efficient processing of CRUD operations with minimal latency.

DevOps Perspective: The DevOps ASRs focus on deployment simplicity, version control, and system reliability, requiring the use of Git for version tracking, support for modular deployment of frontend and backend components, and readiness for future CI/CD integration to ensure smooth updates and maintain system stability.

Logic/Business Layer Perspective: The logic layer ASRs focus on correctness, data integrity, and reusability, requiring centralized business logic within the Service Layer, enforcement of validation rules before database operations, elimination of redundant logic through DRY principles, and structured processing of book inventory data to ensure consistency across the system.

Summary:

This classification ensures that each layer of the system is independently manageable, scalable, and aligned with software engineering principles such as separation of concerns, modularity, and maintainability.